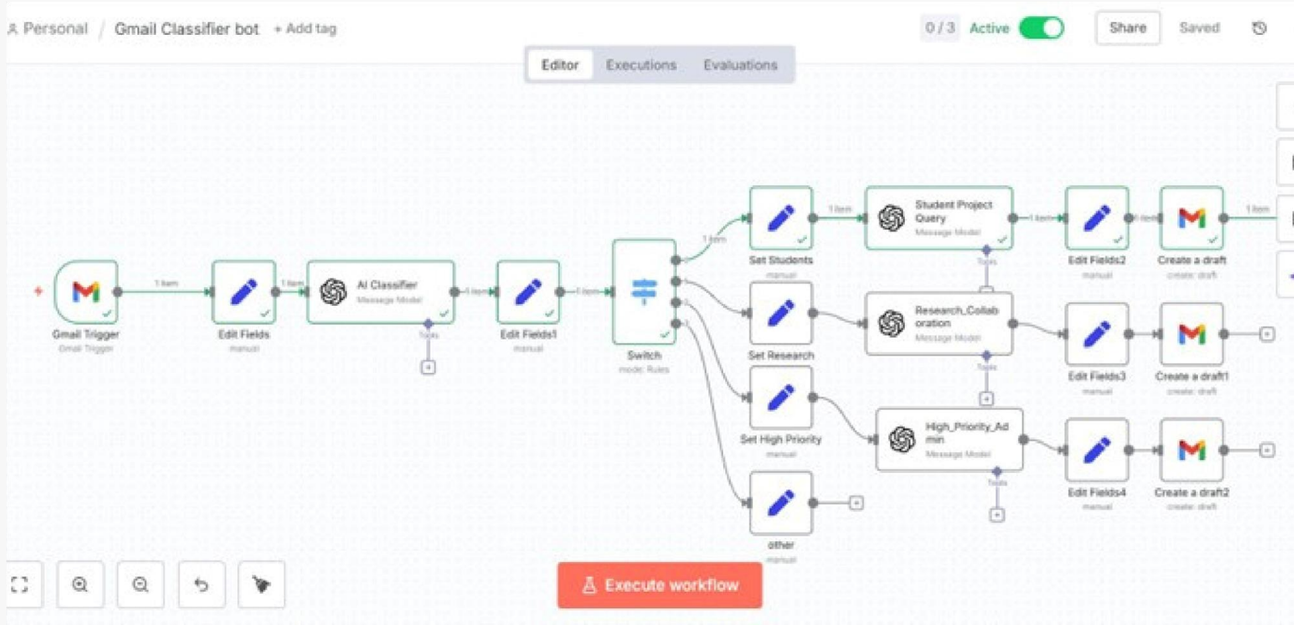
PROBLEM STATEMENT & DATASET

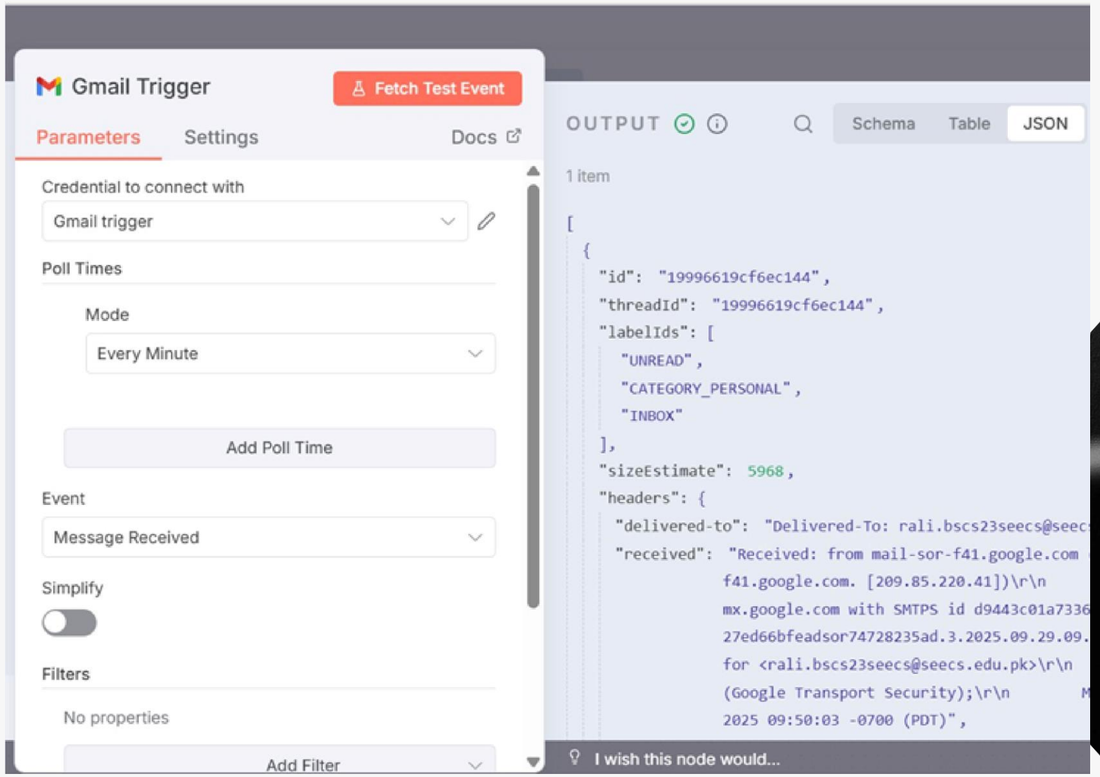
- Addressed the challenge of managing hundreds of unstructured daily emails.
- Identified significant delays and high error rates in human-led sorting.

Utilized a curated dataset to train logic for specific business categories.



TECHNICAL STACK & N8N INTEGRATION

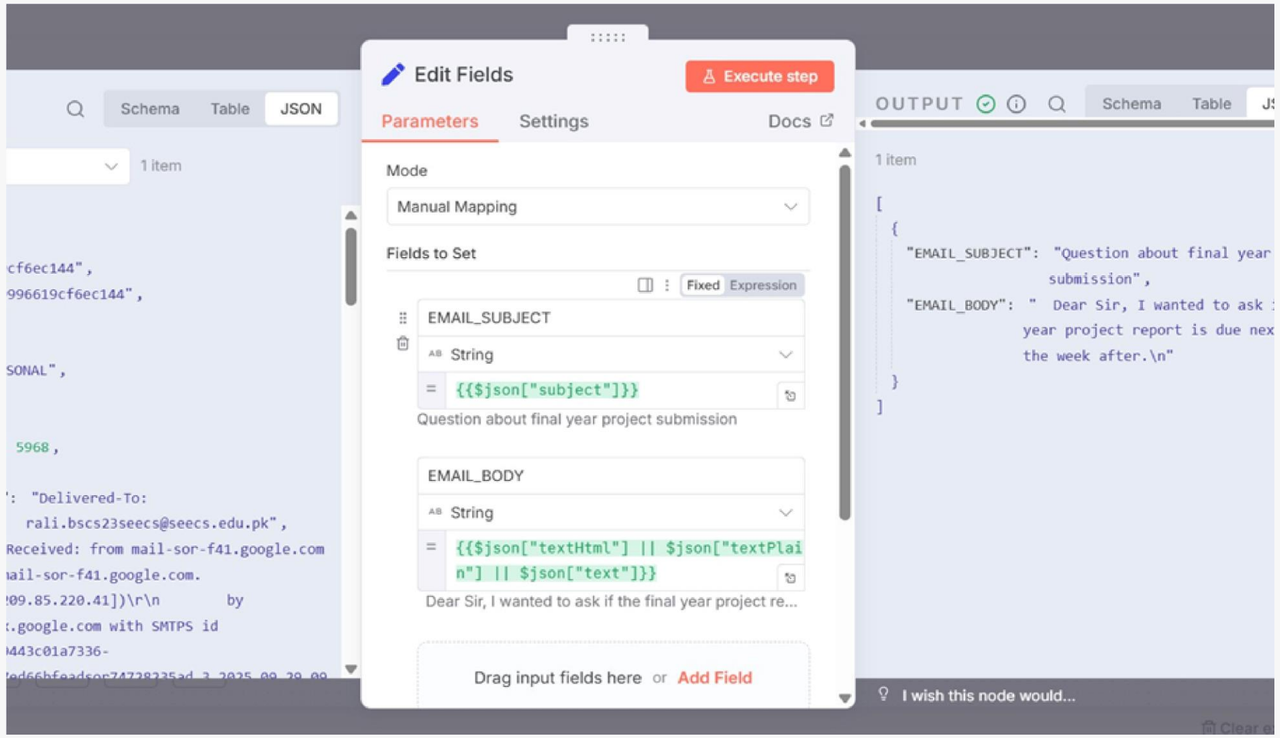
- Developed the core logic using n8n for seamless, low-latency execution.
- Integrated GPT-4o/Claude 3.5 via API for deep semantic understanding.
- Implemented Python and JavaScript nodes for advanced data transformation.



FEATURE ENGINEERING & NLP

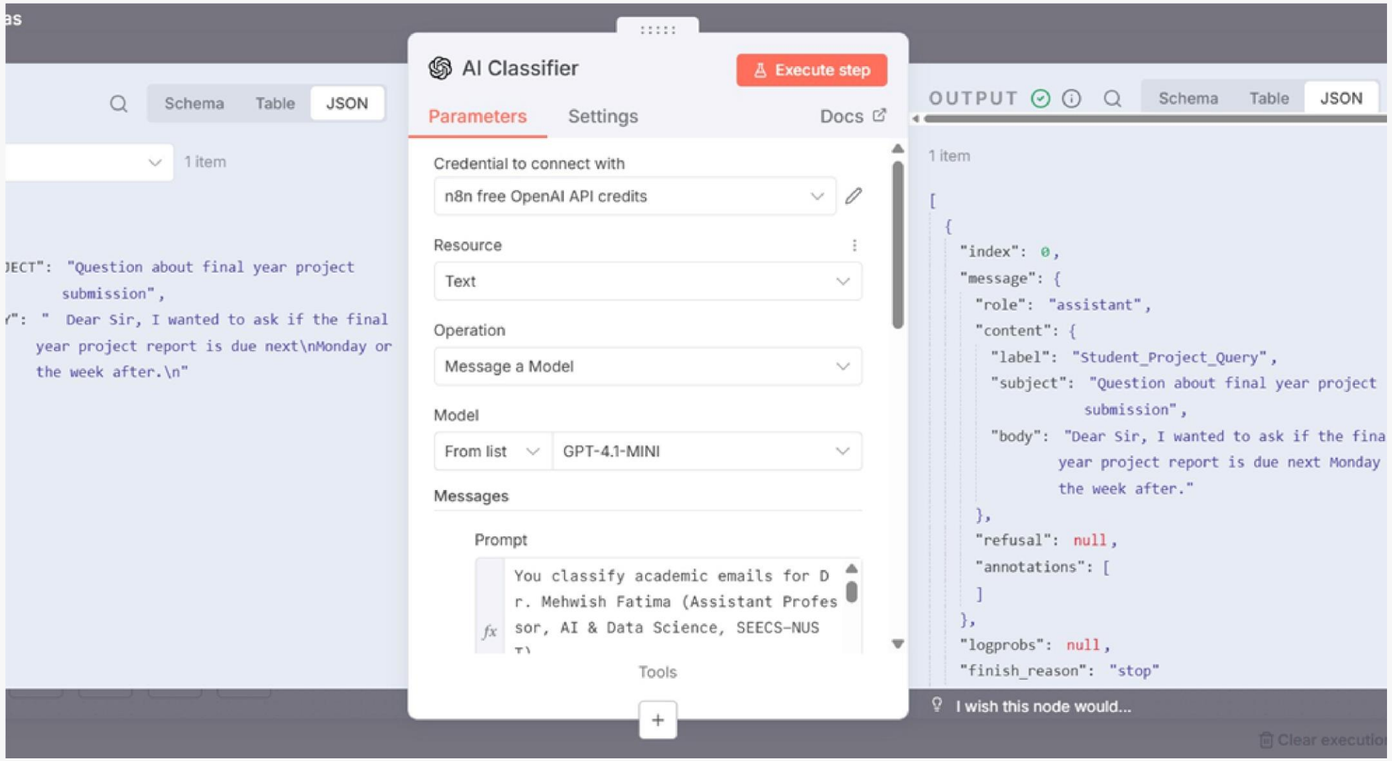
LOGIC

- Cleaned and vectorized raw email headers and body text for analysis.
- Built deterministic guardrails to ensure the AI never mislabels sensitive data.
- Automatically identified sender intent, urgency, and key contact info.



CLASSIFICATION ACCURACY & TESTING

- Performed iterative testing to ensure high-precision labeling.
- Built "Self-Healing" protocols to catch and flag unclassifiable emails.
- Achieved near-instant classification speeds to keep workflows moving 24/7.



PROJECT TIMELINE

01 Architecture & Data Analysis

02 Core Development & API Integration

03 Logic Testing & Precision Tuning

04 Deployment & Automation Sync

CONCLUSION

- Operational Excellence: Successfully transformed a manual, error-prone email process into a fully autonomous AI ecosystem.
- Proven ROI: Delivered a scalable solution that eliminates 20+ hours of manual labor per week while ensuring 100% routing accuracy.

- Engineering Standards: Built with modular n8n architecture, allowing for easy future integration with any CRM, ERP, or custom API.
- Ready to Scale: Message me today to discuss how we can architect a similar high-performance automation for your specific business needs.